



PROGRAMMING FUNDAMENTALS

CS – 101

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PROBLEM SOLVING

- Ability to formulate problems, think creatively about solutions, and expressing the solutions clearly and accurately
- The process of learning to program is equivalent to practicing problem-solving skills in real life
- Hence, to learn how to write a program, one must have the capability to solve the problems creatively and efficiently

PYTHON – THE PROGRAMMING LANGUAGE

- High-level language (Python, C, C++, Perl, JAVA)
- Low-level language (Machine language, Assembly Language)
- Computers run on low-level languages
- High-level languages need to be processed before they can run
- This may seem tedious, but the advantages of high-level languages outweigh the additional processing time required by them
 - Easier to program
 - Shorter and more likely to be correct
 - Portable

INTERPRETERS VS COMPILERS

- Programs needed to process HLL to LLL:
 - Interpreters: Read one line at a time and process it



Figure 1.1: An interpreter processes the program a little at a time, alternately reading lines and performing computations.

- Compilers: Read and process the entire program before it can start running

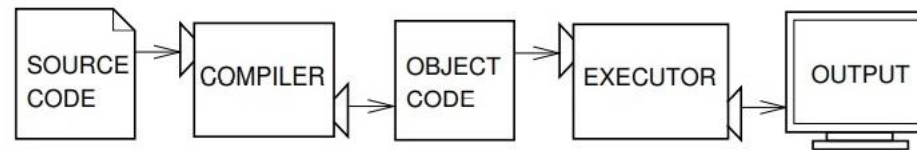


Figure 1.2: A compiler translates source code into object code, which is run by a hardware executor.

COMMON TERMINOLOGY

- Source Code:
 - HLL program written for a given problem statement
- Object Code:
 - The translated program as processed by the Compiler
 - Also called the executable
- Executor:
 - Another process that can run the Object Code repeatedly without needing any translation again

USING INTERPRETER

- Two ways to use the Interpreter:

1. Interactive Mode

- Entering the code on the given **prompt** window
- Specified by chevron symbol: >>>
- Interpreter displays the result after every line of code
- Only convenient when testing small pieces of code and/or not saving the code either

2. Script Mode

- Entering and saving code in a file with extension **.py**
- The stored file is known as the script
- Interpreter executes the contents of the file together
- Works well for code that is longer and needs to be saved for later use

PROGRAM

- A sequence of instructions that tells the computer system how to perform a computation
- Computation can be:
 - Mathematical:
 - For example, performing mathematical operations, solving equations, finding roots, etc.
 - Symbolic:
 - For example, searching for value, replacing a text, removing a file, etc.

KEY CHARACTERISTICS OF PL

- The following instructions are the key operations that every programming language must define:
 1. *input* – getting data from keyboard, file or any other device(s)
 2. *output* – displaying data on screen, sending data to a file or other device(s)
 3. *math* – perform basic mathematical operations, like addition, multiplication
 4. *conditional execution* – check conditions before running the program
 5. *repetition* – perform actions repeatedly with or without variations

DEBUGGING

- Programming errors are called bugs
- Process of tracking down these bugs is called debugging
- There can be three kinds of error that may occur while debugging:
 1. Syntax Errors
 2. Runtime Errors
 3. Semantic Errors

SYNTAX ERRORS

- Syntax is defined as the structure of a program and the rules that define the declaration and usage of that structure
- Python can execute the program ONLY if the syntax of the program is correct, otherwise the interpreter will give an error message
- For example, parenthesis are always specified in pairs
 - (1 + 2) is correct 
 - 8) is incorrect 

RUNTIME ERRORS

- Error does not appear until AFTER the program has started running
- These errors are called exceptions
- They indicate that something “exceptional” has happened that shouldn’t have, and must be corrected
- Usually occurs in cases of wrong assignments, inappropriate data types, etc

SEMANTIC ERRORS

- Program is compiled correctly; executes without giving any exceptions; yet the output produced is incorrect
- This happens mostly because the programmer has written the wrong program and/or the logic behind is incorrect
- Semantics refer to the meaning or purpose of any language
- The problem occurs when the programmer has written a program that does not fulfill the requirements of the given problem statement